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Department of Medicine
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# Chronic Obstructive Pulmonary Disease
# &
# Smoking Cessation

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*Oct, 2023*

# Learning Objectives : You will learn

- the risk factors & pathophysiology of COPD
- the clinical features and complications of COPD
- the investigations & principles of management of COPD
- the psychology of smoking, the methods of and difficulties encountered in smoking cessation
- a clinical case study

# Definition of COPD

- COPD, a common preventable and treatable disease, is characterized by persistent respiratory symptoms and airflow limitation that are due to airway &/or alveolar abnormalities usually caused by significant exposure to noxious particles or gases.

- Exacerbations and comorbidities contribute to the overall severity in individual patients.

# Mechanisms Underlying Airflow Limitation in COPD

## Small Airways Disease

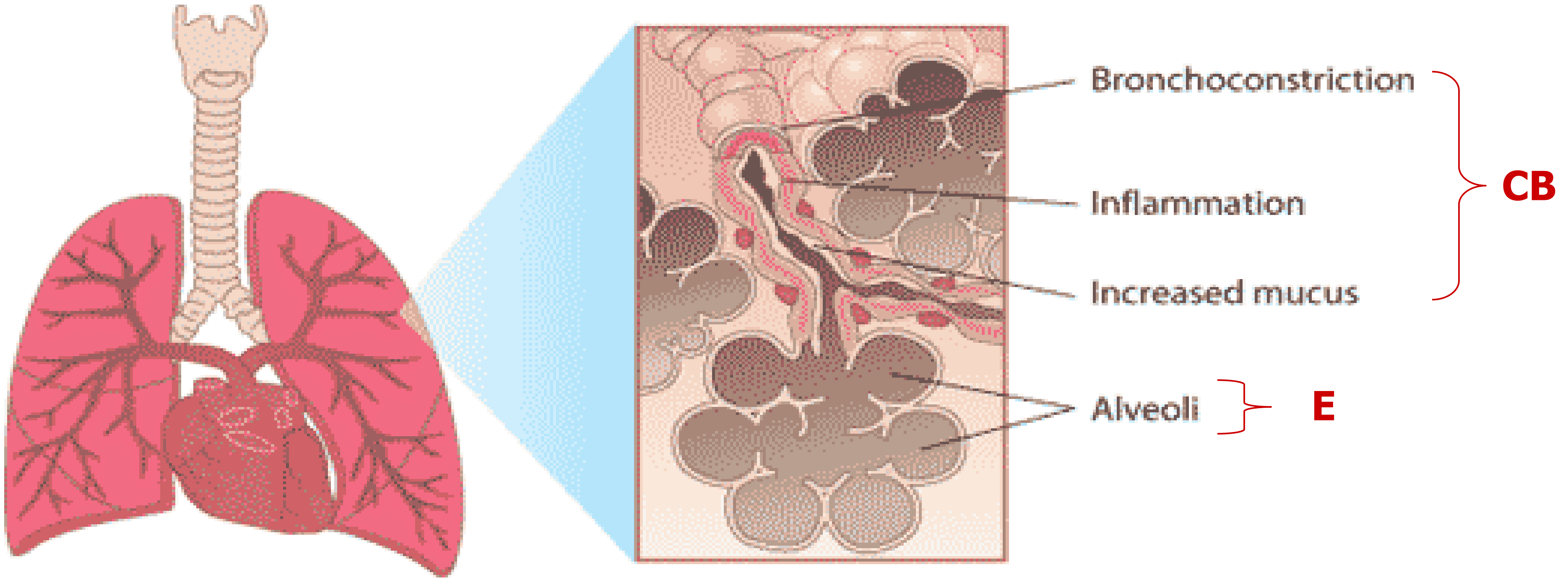
- Airway inflammation
- Airway fibrosis, luminal plugs
- Increased airway resistance

## Parenchymal Destruction

- Loss of alveolar attachments
- Decrease of elastic recoil

**AIRFLOW LIMITATION**

# Chronic Obstructive Pulmonary Disease (COPD)




*CB – chronic bronchitis → chronic cough + sputum*
*E -- emphysema → progressive dyspnoea*

# COPD in Hong Kong

- affects ~10% of Hong Kong elderly > 70 yr old
- overall occupied ~10% public medical bed days

- expected increasing incidence because of smoking and worsening air pollution

→ *a very common medical condition !*

# Risk Factors for COPD

| Environmental Factors | Host Factors |
|---|---|
| * Cigarette Smoking | |
| Air Pollution | * $\alpha$1-antitrypsin deficiency |
| Passive smoking | |
| Indoor biomass combustion, e.g. wood (especially women in rural regions/countries) | |

* *By far the most important factor accounting for > 85% patients*

* *Rare ; considers it when we see a young patient with emphysema (e.g. < 45 yr old, esp Caucasians)*

# COPD : Clinical Presentations

***Typically →***

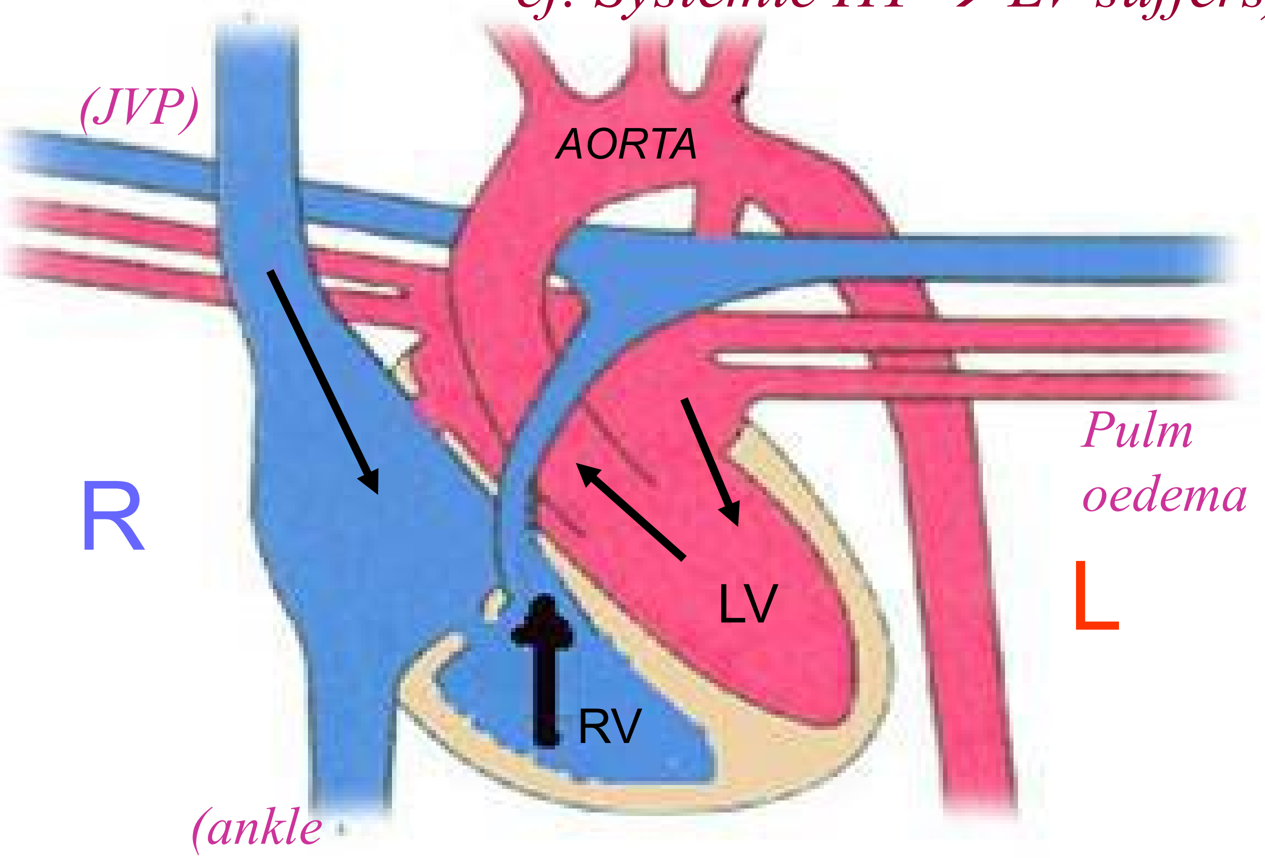
- Middle age / elderly , men > women
- Nearly always Hx of chronic smoking
- Chronic cough and sputum : *years !* (*usually whitish and mucoid unless exacerbations*) – (C.B)
- Progressive shortness of breath – (E)
- Features of Complications → ***hospitalization***

# COMPLICATIONS OF COPD

- Acute exacerbation
  - pneumothorax
  - ↑ airflow obstruction
  - infection
  - respiratory failure
- Chronic respiratory failure
- Cor pulmonale (*chronic hypoxemia → pulmonary HT → RVH & F*)
- Pulmonary thromboembolism

# Concept of Cor Pulmonale

*(Pulmonary HT→ RV suffers ;*
*cf. Systemic HT → LV suffers)*



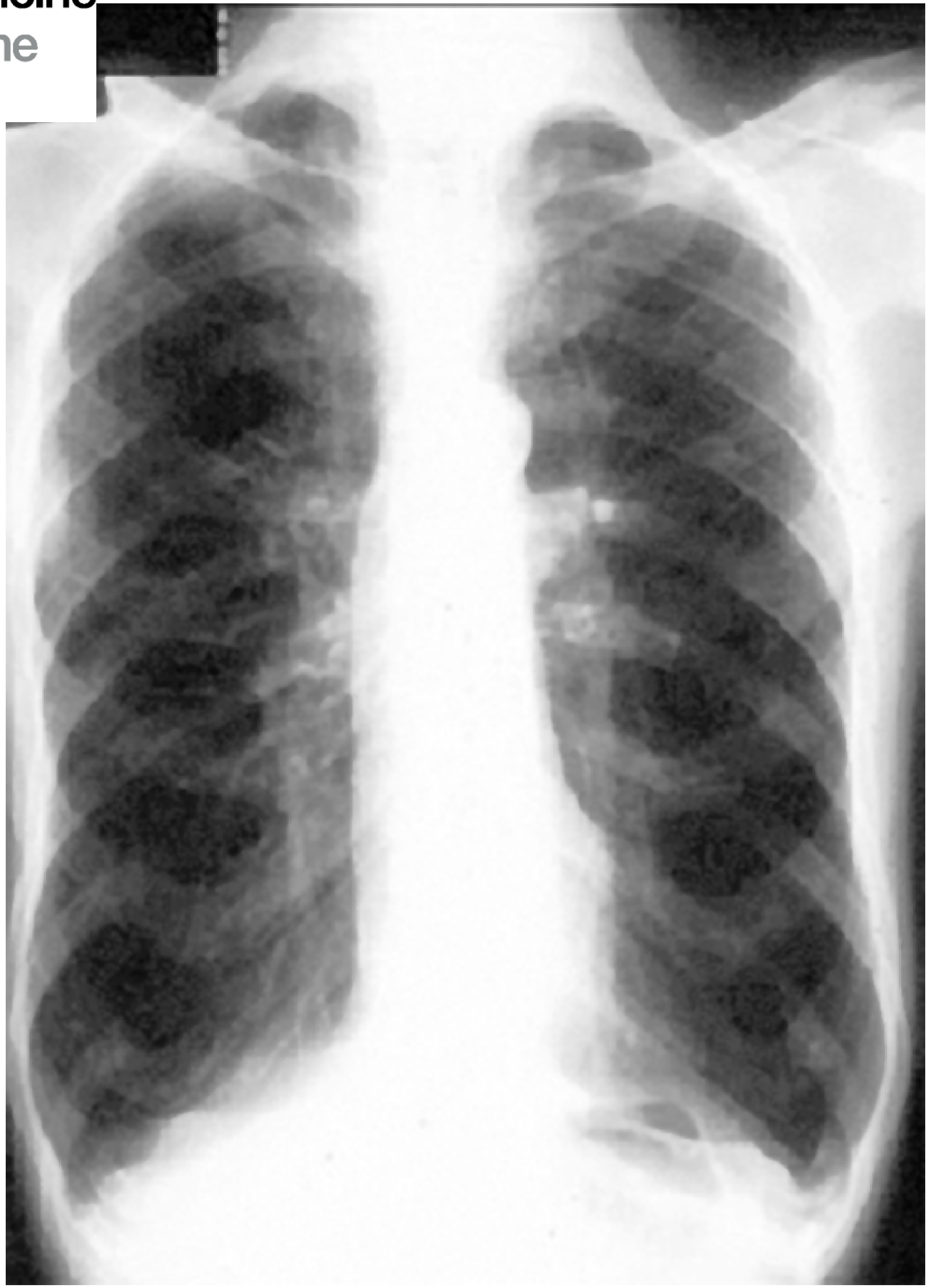

*RVH, RVF*
*(parasternal heave)*

# COPD : Investigations

- **Lung function test –**
  - ***FEV1/FVC < 70% (airflow obstruction)***
  - ***↑RV & TLC ( hyperinflation of lung )***
  - ***↓DLCO ( diffusion is at alveolar level → emphysema with destroyed alveoli)***
- Complete blood counts
- Chest X-ray
- Arterial blood gases (*for resp failure*)
- Sputum examination (*infections*)
- ECG ± echocardiogram (*cor pulmonale*)

. Hyperinflation

. Hypertranslucency



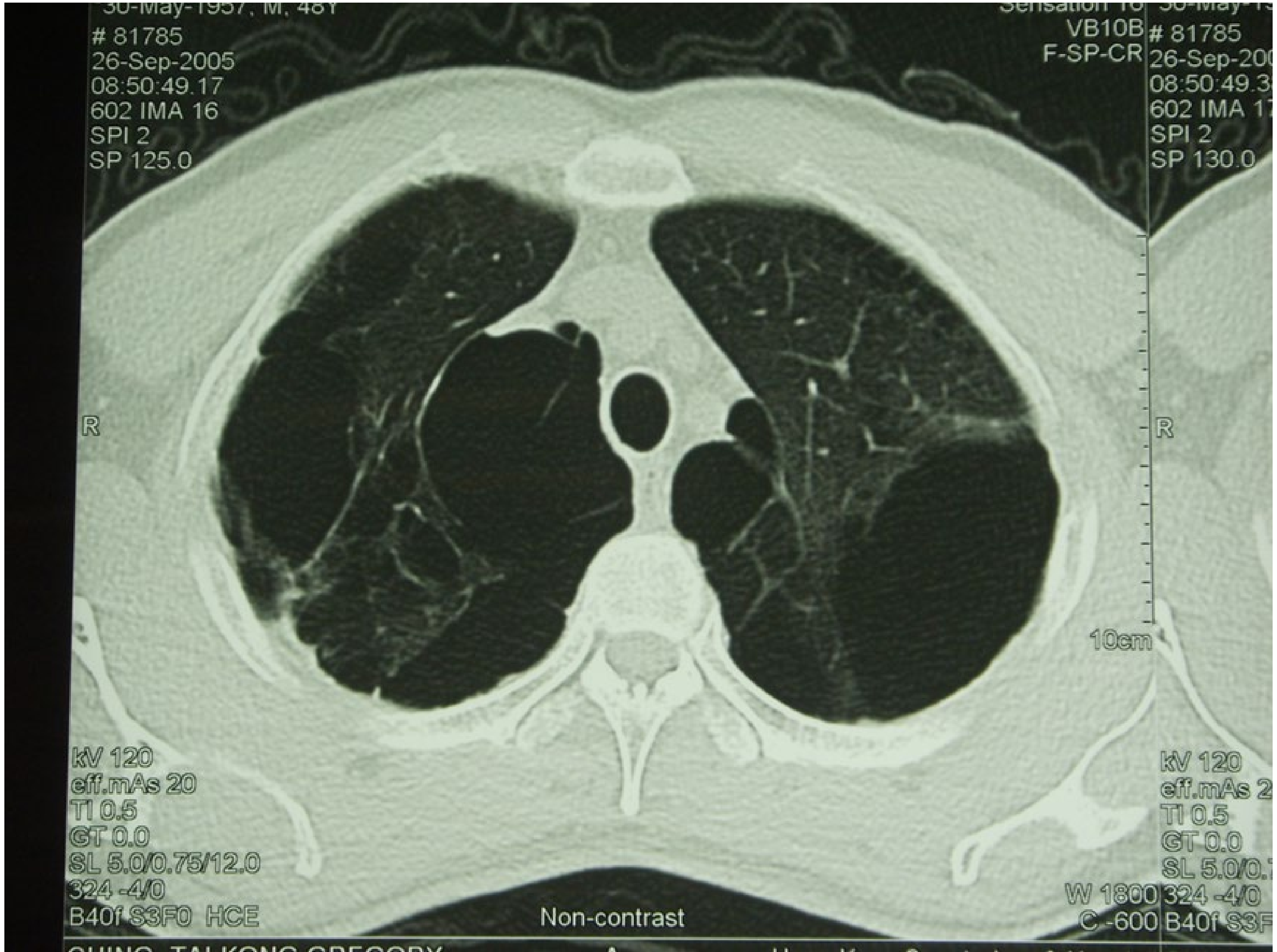
. Hypertranslucency
. Hyperinflation

. Cardiomegaly
. Prominent pulmonary arteries

↓

*COPD complicated by pulmonary hypertension & cor pulmonale*

# Pulmonary emphysema



# Management Program Components

- Assess and monitor disease
- Reduce risk factors
- Manage stable COPD *(OPDs)*
- Manage exacerbations *(hospitals)*
- End stage COPD

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2023 Teaching Slide Set

# GOLD ABE Assessment Tool

Figure 2.3

Spirometrically confirmed diagnosis → Assessment of airflow obstruction → Assessment of symptoms/risk of exacerbations

Post-bronchodilator FEV1/FVC < 0.7

| GRADE | FEV1 (% predicted) |
|---|---|
| GOLD 1 | ≥ 80 |
| GOLD 2 | 50-79 |
| GOLD 3 | 30-49 |
| GOLD 4 | < 30 |

**EXACERBATION HISTORY**
(PER YEAR)

| EXACERBATION HISTORY | mMRC 0-1<br>CAT < 10 | mMRC ≥ 2<br>CAT ≥ 10 |
|---|---|---|
| ≥ 2 moderate exacerbations or ≥ 1 leading to hospitalization | E | E |
| 0 or 1 moderate exacerbations (not leading to hospitalization) | A | B |

**SYMPTOMS**



GLOBAL INITIATIVE FOR CHRONIC OBSTRUCTIVE LUNG DISEASE™

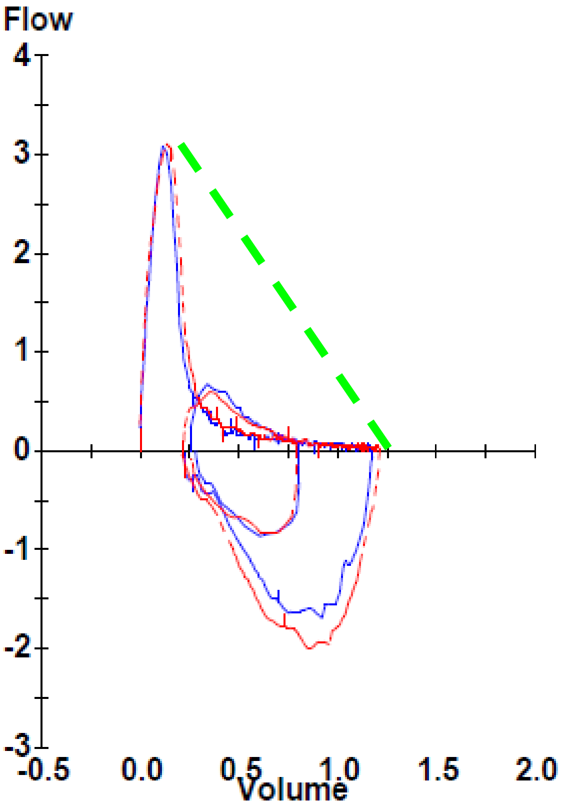
# Very severe airflow limitation

**Best Data**

| | | Ref | Pre | % Ref | Post | % Ref | %Chg |
|---|---|---|---|---|---|---|---|
| **Spirometry** | | | | | | | |
| FVC | Liters | 2.53 | 1.18 | 47 | 1.22 | 48 | 3 |
| FEV1 | Liters | 1.87 | 0.47 | 25 | 0.50 | 27 | 8 |
| FEV1/FVC | % | 74 | 39 | | 41 | | |
| FEV3 | Liters | 2.53 | 0.71 | 28 | 0.76 | 30 | 7 |
| FEF25-75% | L/sec | 2.48 | 0.12 | 5 | 0.12 | 5 | 5 |
| FEF50% | L/sec | 3.19 | 0.11 | 3 | 0.14 | 4 | 22 |
| FEF75% | L/sec | 1.05 | 0.09 | 8 | 0.04 | 4 | -56 |
| PEF | L/sec | 5.50 | 3.09 | 56 | 3.11 | 56 | 1 |
| FIVC | Liters | 2.53 | 0.95 | 38 | 1.00 | 39 | 5 |
| FIF50% | L/sec | | 1.27 | | 1.55 | | 23 |
| **Lung Volumes** | | | | | | | |
| TLC | Liters | 3.58 | 2.00 | 56 | | | |
| VC | Liters | 2.81 | 1.18 | 42 | | | |
| IC | Liters | | 0.85 | | | | |
| FRC N2 | Liters | 1.92 | 1.15 | 60 | | | |
| ERV | Liters | | 0.03 | | | | |
| RV | Liters | 1.21 | 0.82 | 68 | | | |
| RV/TLC | % | 34 | 41 | | | | |
| Wash Time | Min | | 3.0 | | | | |

Diffusing Capacity (Hb 13.5)

| | | | | |
|---|---|---|---|---|
| DLCO | mmol/kPa.min | 4.6 | 1.5 | 32 |
| DL Adj | mmol/kPa.min | 4.6 | 1.5 | 32 |
| DLCO/VA | DLCO/L | 1.28 | 1.22 | 95 |
| Kroghs K | 1/min | 3.54 | 3.14 | 89 |
| VA | Liters | | 1.21 | |
| IVC | Liters | | 0.89 | |
| FI CH4 | % | | 0.300 | |
| FE CH4 | % | | 0.150 | |
| FI CO | % | | 0.300 | |
| FE CO | % | | 0.075 | |
| BHT | Sec | | 13.20 | |



# Management Program Components

- Assess and monitor disease
- Reduce risk factors：*stop smoking*
- Manage stable COPD . *Bronchodilator (BD) , Inhaled Corticosteroids( ICS)*
  *. Long-term Oxygen Therapy (LTOT)*
  *. Rehabilitation*
- Manage exacerbation . *Controlled oxygen therapy*
  *. Antibiotics*
  *. Systemic steroid*
  *. Non-invasive Ventilation (NIV)*
- End stage COPD. *Lung Volume Reduction Procedures / Lung Transplantation*

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SYMPTOMS

## Initial Pharmacological Treatment

Figure 4.2

| | | |
|---|---|---|
| ≥ 2 moderate exacerbations or ≥ 1 leading to hospitalization | **GROUP E** <br> **LABA + LAMA*** <br> *consider LABA+LAMA+ICS* if blood eos ≥ 300* | |
| 0 or 1 moderate exacerbations (not leading to hospital admission) | **GROUP A** <br> **A bronchodilator** | **GROUP B** <br> **LABA + LAMA*** |
| | mMRC 0-1, CAT < 10 | mMRC ≥ 2, CAT ≥ 10 |

*single inhaler therapy may be more convenient and effective than multiple inhalers
Exacerbations refers to the number of exacerbations per year

# COPD Management : Principles (1)

- Stopping smoking
- Bronchodilators (BD)
  - Anti-cholinergics
  - $\beta_2$ agonists

*Inhaled preferred*

- Anti-inflammatory Rx :
  - Inhaled corticosteroids (ICS)
  - New oral phosphodiesterase inhibitor (Roflumilast)

*When frequent exacerbations*



**$\beta_2$ agonists** – **salbutamol (*Ventolin*), terbutaline (*Bricanyl*)**
**salmeterol (*Serevent*), formoterol (*Oxis*)**

**Anti-cholinergics** **ipratropium (*Atrovent*) – tiotropium (*Spiriva*)**

**Anti-inflammatory ---**

**Inhaled corticosteroids --**

- **beclomethasone (*Becotide, Becloforte*)**
- **budesonide (*Pulmicort*)**
- **fluticasone (*Flixotide*)**

**Roflumilast (*Daxas*) --- *a new oral non-steroid anti-inflammatory therapy specific for COPD***

# Management Programme Components

- Assess and monitor disease
- Reduce risk factors
- Manage stable COPD *(OPDs)*
- Manage complications *(hospitals)*
- End stage COPD

# COPD Management : Principles (2)

**Management of complications** – *which account for most of the hospitalizations & mortality*

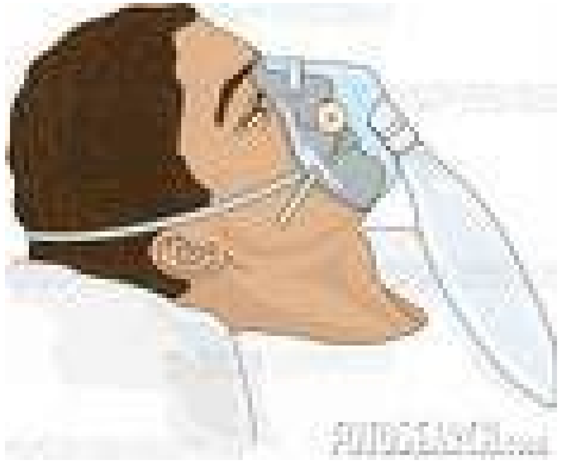
- *Acute exacerbation*
  - controlled $O_2$ therapy
  - exclude and treat pneumothorax
  - systemic corticosteroid
  - ↑ inhaled bronchodilators
  - antibiotics for infections
  - non-invasive ventilation for decompensated type II respiratory failure
- *Cor pulmonale*
  - diuretics
  - salt & fluid restriction

# Controlled $O_2$ Therapy

- **Aim** - to use $O_2$ in a controlled manner to achieve a target $PaO_2$ of 8kPa ($SaO_2$ > 90%) without significant $CO_2$ retention or acidosis
- **Modes**
  - nasal cannula (1-2 L/min)
  - Venturi mask (24%, 28%)
- **Monitor**
  - clinical (conscious levels, blood pressure, pulses)
  - pulse oximetry
  - arterial blood gas measurement

## Nasal Cannula

## Oxygen Mask



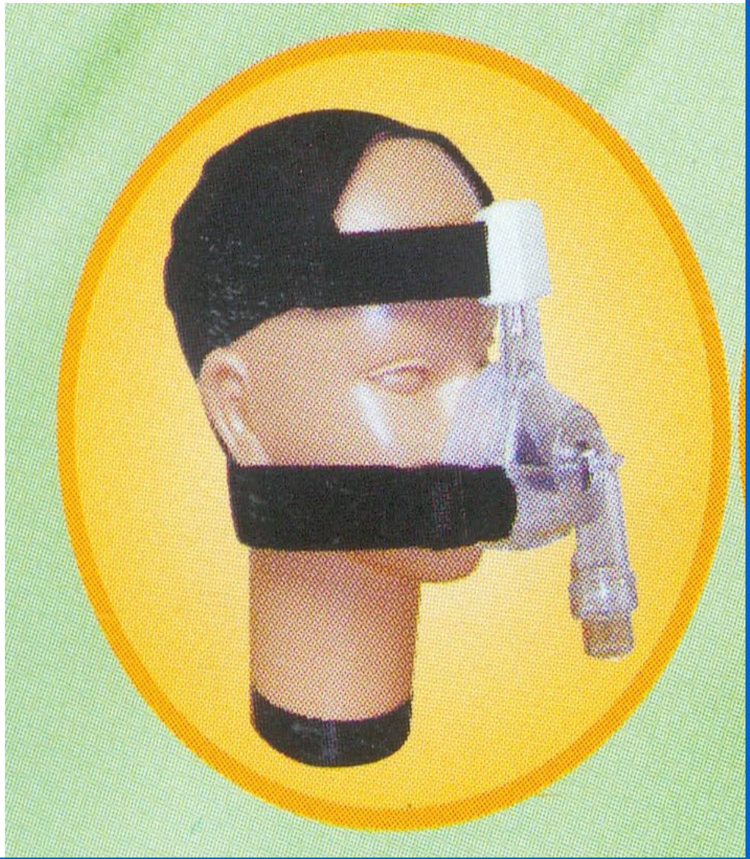
# Antibiotics in COPD (1)

- Commonest sputum bacterial isolates :
  - *Haemophilus influenzae*
  - *Streptococcus pneumoniae*
  - *Moraxella catarrhalis*
  - *(Pseudomonas aeruginosa)*
- Choice of empirical antibiotics is usually clinical, basing on knowledge of local bacterial & resistance pattern

# Antibiotics in COPD (2)

- indicated when ≥ 2 of the following are present:
  - ↑ dyspnoea
  - ↑ sputum volume
  - purulence of sputum
- inexpensive antibiotics are adequate in most cases : amoxycillin, augmentin, macrolide, cephalosporin

# NIV for COPD Type II Respiratory Failure



Head Gear / Head strap
for NIV using
face mask /nasal mask

→ *tightly fit to minimize air leak*

→ *no intubation, hence "non-invasive"*

→ *intubation & IPPV when NIV fails*


Quantum PSV
0.575
Prop up

# COPD Management : Principles (3)

## Long Term Management

- Long term home $O_2$ therapy (LTOT)
  - $O_2$ concentrator
  - improves survival
- Pulmonary Rehabilitation Programme + flu vaccine

## End-Stage COPD

- Lung volume reduction surgery *(LVRS)*
- Lung transplantation

# Criteria for LTOT in COPD

1. Resting $PaO_2$ < 7.3 kPa by two separate measurements when breathing room air in stable condition on optimal medical treatment

2. Resting $PaO_2$ < 8 kPa *( i.e. relaxing the criteria)* when in addition the patient has :
   . Secondary polycythaemia (haematocrit > 55%)
   . *Cor pulmonale*
   . Pulmonary hypertension
   . Nocturnal hypoxaemia

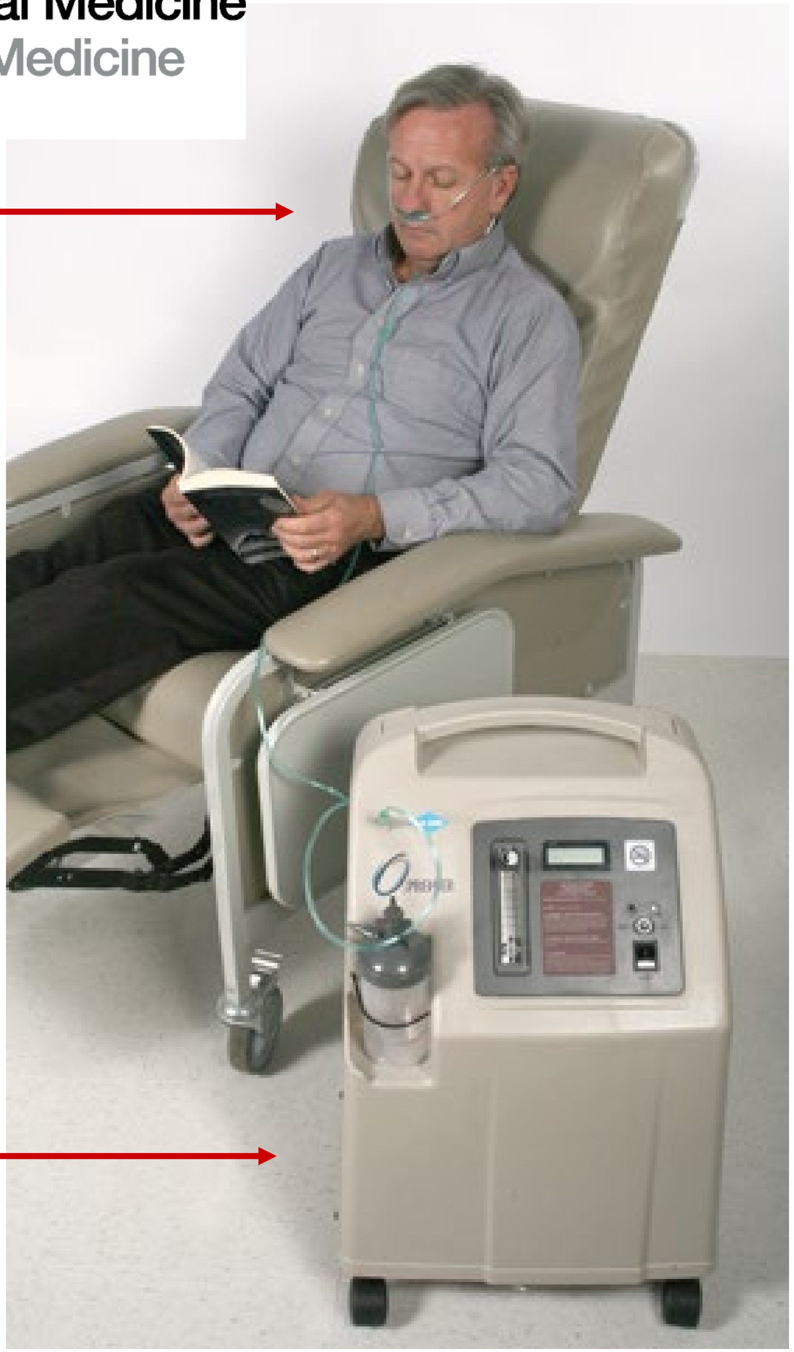
# $O_2$ Concentrators for Home LTOT

. An electrical appliance

. A molecular sieve that takes and concentrates $O_2$ from air → non-stop supply

. Patient can choose to buy or to rent

Nasal cannula




$O_2$
concentrator

. Goal : $SaO_2 > 90\%$

. may ↑ flow in
exercise & sleep

# Portable $O_2$ concentrators now available

- 1 - 2 kg in weight
- With batteries working for 3 hours ( bigger size → up to 10 hour )
- Enables patients to travel a bit

# Rehabilitation

A multidisciplinary pulmonary rehabilitation programme may include:

- Physiotherapy
- Muscle/exercise training
- Nutritional support -- for cachexia
- Psychotherapy
- Education
- Ventilatory assistance
- Home care

# Pulmonary Rehabilatation Class -- exercises

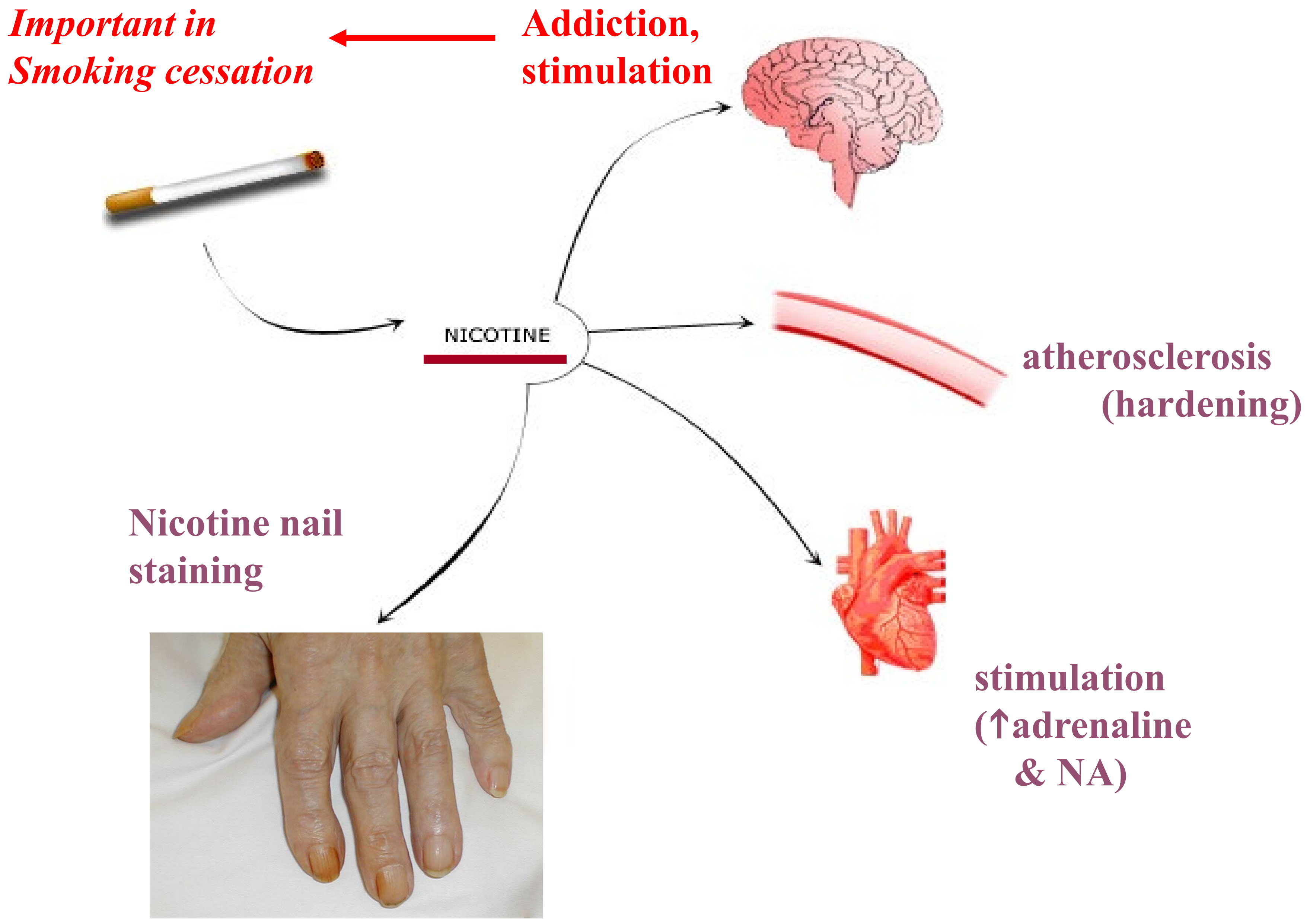
# Exercise Training

- Improves muscle function and increases exercise tolerance
- Reduces dyspnoea
- Improves quality of life.

*NEJM 2009;360:1329-1335*
*Casaburi R & ZuWallack R,*

# Smoking Cessation

- > 4,000 chemicals
- > 40 carcinogens
- Nicotine
  (→ addiction
  → brain)
- CO (→ heart)


*Important in Smoking cessation*
Addiction, stimulation
NICOTINE
atherosclerosis (hardening)
Nicotine nail staining
stimulation (↑adrenaline & NA)

# SMOKING CESSATION (1)

## Why smoke?

- overlearnt habit
- association and secondary reinforcement:
  intertwined with the daily life events: fun, grown-up or male image, boredom
- routine
- craving
- stress
- social-peer pressure
- relaxation

NB. Nicotin is a stimulant and addictive.

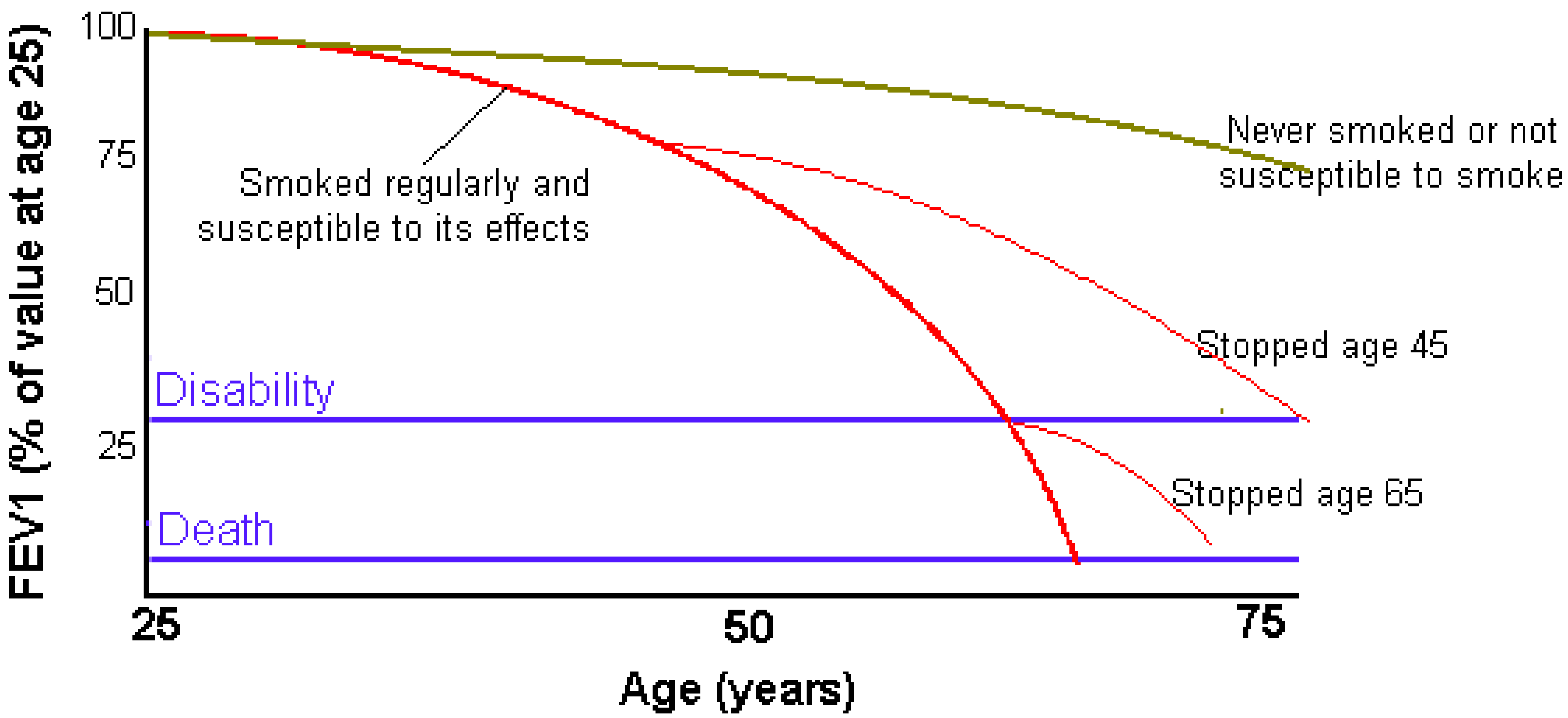
# SMOKING CESSATION (2)

**Why give up?**

*Need to find a personal reason to quit smoking*

- save money
- more acceptable socially
- not harm others
- improve health - yourself and family
- clothes and home smell fresher
- increased appreciation of taste, smell
- fire hazards

# Fletcher's graph: Decline of $FEV_1$ with age and smoking habits



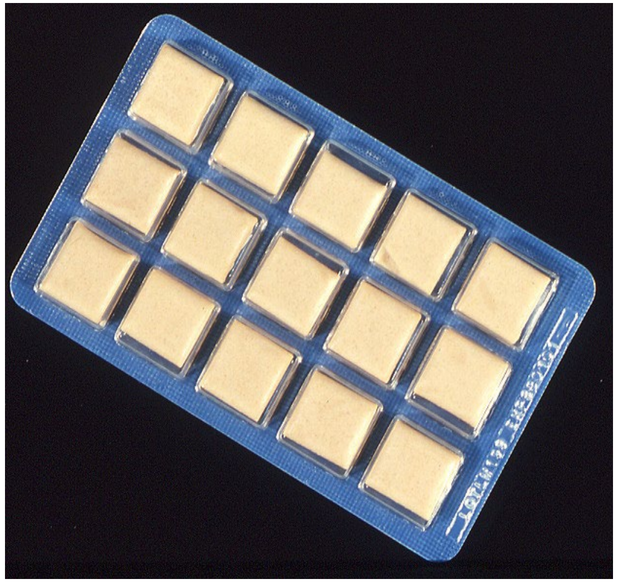

**It's never too late to quit**

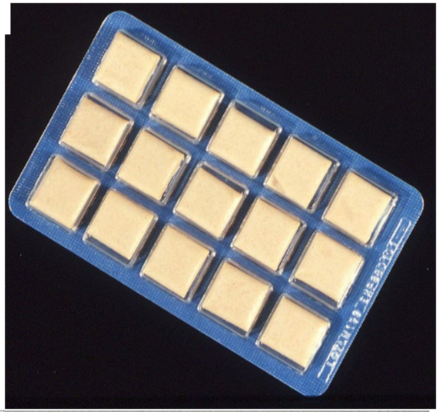
# SMOKING CESSATION (4)

## Withdrawal Symptoms

- craving
- coughing
- hunger and weight gain
- bowel disturbance
- sleep disturbance
- dizziness and paraesthesia
- mood swings, lack of concentration and irritability

# Nicotine : gums & patches





1. Nicotine - gum, patch, inhaler

2. Bupropion SR *(Zyban/Wellbutrin SR)* –
antidepressant with specific
use in smoking cessation

3. Varenicline (*Champix*)
acts at the same receptor in the brain
as nicotine (ACh receptor) → dual action :
. ↓ withdrawal S/S when quitting
. blocks the reinforcing effects of nicotine
when smoking

*Plus*
- health care professional counseling
- social support by family & friends

***Success rate at 1 year***
***→ only about 20–35%!***
***Efficacy : Varenicline : best***

# As medical students and future doctors, please :

- Learn more about smoking and health
- Do Not smoke
- Ask your parents, friends and patients not to smoke
- Champion anti-smoking campaigns

*www.who.int/tobacco*
*www.cdc.gov/tobacco*
*www.surgeongeneral.gov/library/secondhandsmoke*
*www.tobaccocontrol.com (journal)*

# COPD and Comorbidities

**Some common comorbidities occurring in patients with COPD with stable disease include:**

- Cardiovascular disease (CVD)
- Heart failure
- Ischaemic heart disease (IHD)
- Arrhythmias
- Peripheral vascular disease
- Hypertension
- Osteoporosis
- Anxiety and depression
- COPD and lung cancer
- Metabolic syndrome and diabetes
- Gastroesophageal reflux (GERD)
- Bronchiectasis
- Obstructive sleep apnea

# Learning Objectives : You have learnt

- the risk factors & pathophysiology of COPD
- the clinical features and complications of COPD
- the investigations & principles of management of COPD
- the psychology of smoking, the methods of and difficulties encountered in smoking cessation

# TELL US WHAT

JC Whole Class Session

# YOU THINK

# THANK YOU!

JC
WCS

Look for today's teacher
and the others you may have missed.
(HKU Portal login required)